

INTRODUCTION TO DATA ANALYSIS

USING EXCEL

Data Analysis: It is a process of examining, cleaning, transforming & modeling data to discover useful information, patterns, trends to support decision-making.

→ It is turning raw data to useful data.

→ It involves:

① **Data Collection:** gathering the data from relevant sources.

② **Data Cleaning:** Removing errors, handling missing values

③ **Interpreting the results:** explaining why the result occurred & what they mean in real world.

⇒ **Data Analysis & Data Science:** Data Analysis is about understanding past & present data to make informed decisions.

→ **Data Science** is about using data to create systems that can predict, learn & automate decisions.

Data Analysis

Data Science

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| ① Examining the data to find patterns & insights. | ① A broad field involving extraction of knowledge & predictions from data. |
| ② Primary goal is to understand & interpret existing data | ② Primary goal is to build models make predictions & automate decision making. |
| ③ Focuses on what happened & why | ③ Focuses on what will happen or what to do next. |
| ④ These are structured data | ④ These are both structured & unstructured data |
| ⑤ Process involved are: Data cleaning, summarization & visualization. | ⑤ Process involved are: data collection, cleaning, modeling, machine learning, deployment. |

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| ⑦ Tools used: Excel, SQL, Python. | ⑦ Tools used: Python, R, Spark, Big Data Tools. |
| ⑧ Techniques: Descriptive statistics, charts, pivot tables. | ⑧ Machine learning, statistical modeling, AI. |
| ⑨ Outcomes: Reports, dashboard, summaries. | ⑨ Outcomes: Predictive models, intelligent systems. |
| ⑩ Generally easier to learn foundation for Data Science. | ⑩ Requires deeper knowledge of programming, statistics & ML. |

* Types of Data Analysis:

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|---------------|-------------------------|------------------------|-----------------------|-----------------|
| ① Data mining | ② Business Intelligence | ③ Statistical Analysis | ④ Predictive Analysis | ⑤ Text Analysis |
|---------------|-------------------------|------------------------|-----------------------|-----------------|

① Data Mining:

- Process of analyzing large datasets to find hidden patterns, relationships & useful information.
- Focuses on discovering meaningful patterns that were not previously known.
- Patterns found can help in summarizing data, improving decision-making, making accurate predictions.
- Widely used in decision support systems to assist businesses in planning & forecasting.
- Used in industries like finance, healthcare, retail & telecommunications.

② Business Intelligence:

- BI techniques & tools are used for acquisition & transformation of large amounts of unstructured business data to help identify, develop & create new strategic business opportunities.
- Goal is to allow easy interpretation of large volumes of data to identify new opportunities.
- It helps in implementing an effective strategy based on insights that can provide businesses with competitive

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market advantage & long term stability.

③ Statistical Analysis:

→ It is study of collection, analysis, interpretation, presentation & organization of data.

2 types:

(i) **Descriptive Statistics:** Data from entire population or a sample is summarized with numerical descriptors such as mean, standard deviation for continuous data, frequency, % for categorical data.

(ii) **Inferential Statistics:** It uses pattern in the sample data to draw inferences about the represented population or accounting for randomness.

⇒ these inferences can be:

- answering yes/no question (Hypothesis testing)
- estimating numerical characteristics (estimation)
- describing association with data (correlation)
- modeling relationship with data (regression analysis)

④ Predictive Analysis:

- Process of using data to forecast future outcomes.
- Process uses data analysis, machine learning, artificial intelligence & statistical models to find patterns that might predict future behavior.

2 Types:

(i) **Classification models:** attempts to put data objects into one category or another.

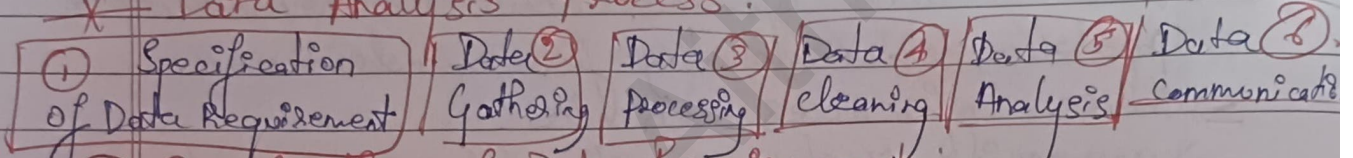
(ii) **Regression models:** try to predict continuous data such as how much revenue that customer will generate during their relationship with company.

⑤ Text Analysis:

- Process of transforming unstructured text documents into usable, structured data.

- It works by breaking apart sentences & phrases into their components. & then evaluating each part's role & meaning using complex software rules & machine learning algorithms.
- It forms the foundation of numerous natural languages processing features including named entity recognition, categorization & sentiment analysis.
- Also known as Text Mining / Text Data Mining which is process of deriving high-quality information from text.

X Data Analysis Process:



① Specification of Data Requirement:

- Figure out what data is needed based on company/business requirements or questions.
- If we want to study group of people. (population)
- Specific variables regarding a population may be specified & obtained
- Data may be numerical / categorical.

② Data Gathering:

- It means collecting the raw information you need to answer a specific question or solve a problem.
- Data can be collected from:
 - Internal Sources: Inside the organization
Ex: CRM software, emp records, Sales data
 - External sources: outside the organization
Ex: Govt websites, online survey, Social media etc.
- Data collected are often unstructured & may include irrelevant information.
- The collected data is later processed & cleaned.

③ Data Processing:

- Data collected must be processed/organised for analysis.
- It includes structuring the data as required for the relevant analysis tools.

④ Data Cleaning:

- Clean the data to prepare it for analysis.
- Involves removing duplicate & anomalous data, reconciling inconsistencies, standardizing data structure & format & dealing with white spaces & other syntax errors.

⑤ Data Analysis:

- By manipulating the data using various data analysis techniques & tools you can find trends, correlations outliers & variations that tell a story.
- You may use data mining to discover patterns within databases / data visualization software to help transform data into an easy-to-understand graphical format.
- Data Visualization may also be used to examine the data in graphical format, to obtain additional insight regarding the messages within the data.
- Statistical Data Models such as Correlation, Regression Analysis can be used to identify the relations among the data variables.

⑥ Data Communication:

- The results of the data analysis are to be reported in a format as required by the users to support their decisions & further action. The feedback from the users might result in additional analysis.
- Can use tables & charts which help in communicating the message clearly & efficiently to the users.

* Microsoft Excel is a powerful electronic spreadsheet program you can use to automate accounting work, organize data & perform wide variety of tasks.

* Spreadsheet Terms:

- ① Quick Access Toolbar
- ② Search Bar
- ③ Name Box.
- ④ Cell (A1)
↓
row.
↓
column
- ⑤ Title Bar
- ⑥ File Tab.
- ⑦ Range.
- ⑧ Status Bar.
- ⑨ New Sheet
- ⑩ Ribbon
- ⑪ Formula Bar
- ⑫ Worksheet
- ⑬ View option.

* Ribbons:

→ It is row of tabs & icons at the top of the Excel window that allows you to quickly find, understand & use commands for completing a certain task.

→ Ribbon in Excel made up of 4 basic components: Tabs, ~~Groups~~ Groups, Dialog Launchers & Command Buttons.

① Ribbon Tab: contains multiple commands logically sub-divided into groups.

② Ribbon Groups: is a set of closely related commands normally performed as part of a larger task.

③ Dialog Launcher: is a small arrow in the lower-right corner of a group that brings up more related commands. Dialog launcher appears in groups that contain more commands than available space.

④ Command button: is the button you click to perform a particular action.

* Range: It refers to a group of one or more selected cells in a worksheet.

→ It can be vertical, horizontal or rectangular.

⇒ Cell addressing:

Relative: =A1+B1 copied to next row become =A2+B2

→ Automatic. =B2/B6

Absolute: \$B\$6 will lock address. =B2/\$B\$6

Mixed: =A\$1 → column (A) fixed & vice versa row can change (A\$1.)

* **Range Names:** When cell / range names become tedious to memorize in a spreadsheet, use the Named Ranges features.

→ It provides a good method of referring to cell addresses.

* **Guidelines for naming ranges:**

- ① 1st character is a letter / underscore (-), naming can be letter, -, no periods &
- ② Names cannot be same as cell reference like P499, Z\$100, R1C1.
- ③ spaces are not allowed.
- ④ Name can contain upto 255 characters.
- ⑤ Names are not case sensitive.

* **Table:** It is a rectangular range of structured data.

- Each row in table corresponds to a single record of the data.
- Each column contains a specific piece of information.
- Each entry in the top row is referred to as column header.
- Select the Range of cells that you want to include in the Table.

* **Tables**

- ① More structured way of working with data.
- ② Range can be converted into a table.
- ③ Provides table tools in ribbon.
- ④ Automatically adds filter buttons to each column header.
- ⑤ Table headers replace column letters when header row disappears.

Ranges.

- ① Simple group of cells without structure.
- ② Do not automatically become a table.
- ③ No special tools only normal Excel options.
- ④ No automatic filters added manually.
- ⑤ Column letters remain headers do not replace them.

- ⑥ A formula in one cell of a column automatically fills entire column.
- ⑦ Can use table name, column header names.
- ⑧ Can extend table using the triangular control at bottom right corner.
- ⑨ Supports sliders for filtering data.
- ⑥ Formula must be copied manually.
- ⑦ Must use cell references unless named ranges used.
- ⑧ Must manually insert rows/columns.
- ⑨ Do not support Slicers.

* Basics Of Formula: Equal sign (=) followed by one or more operands, which can be values, cell references, ranges, range names or function names — separated by one or more operators — which are symbols that combine the operands in some way, such as the plus sign (+) & the greater than sign (>).

⇒ Ex: $=10+5$ (arithmetic) $=10<5$ (comparison)

→ Preference formula operators:

- ① [Colon] — Range — provides range from 2 cell references. Ex: A1:C5.
- ② (space) B2:E8 — Intersection range that is intersection of 2 ranges.
- ③ (comma) — union — range that is union of two ranges. Ex: A1:C5, B2:E8.

* Excel Functions:

Arithmetic, Statistical & Logical Functions.

- ① Arithmetic:
 - ① Sum (no1, no2, ... or range)
 - ② SumIF (range, criteria, sum-range)
 - ③ SumPRODUCT
 - ④ Round
 - ⑤ Product
 - ⑥ Power
 - ⑦ GCD
 - ⑧ Lcm
 - ⑨ SumIFS
 - ⑩ SumSQ etc.

* Statistical Functions:

① **COUNT**: count no of cells containing a number.

Syntax: =COUNT (value1, [value2], ...).

② **COUNTA**: will count the number of cell containing any kind of information, including numbers, errors, values & empty text.

Syntax: =COUNTA (value1, [value2], ...).

③ **COUNTBLANK**: only count blank/empty cell.

Syntax: =COUNTBLANK (range).

④ **COUNTIFS**: works on one or more condition in a given range & counts the cell that meets the condition.

Syntax: =COUNTIFS (range1, criteria1, [range2], [criteria2], ...).

⑤ **MAX**: maximum/largest numeric value within given set of data/array.

Syntax: =MAX (number1, [number2], ...).

⑥ **LARGE**: Similar to MAX but it returns the n^{th} largest value within the given data or array set.

Syntax: =LARGE (array, k)

⇒ Other: COUNTIF, SUMIF, SUMIFS, AVERAGE, AVERAGEIF, MEDIAN, MODE etc.

* Logical Functions:

① **AND**: =AND (A2 > 1, A2 < 100).

② **NOT**: =NOT (A2 > 100)

④ **OR**: =OR (A2 > 1, A2 < 100)

⑤ **IFS**: ⑥ **IF** ⑦ **XOR** etc.

Text Functions:

LEFT, RIGHT, MID, TRIM, LEN, FIND, SEARCH, REPT, CONCATENATE, REPLACE, VALUE, UPPER, LOWER, PROPER, CHAR, CLEAN, SUBSTITUTE.

→ Proper trim spaces & proper case.

Ex: ~~Ram~~ "I am seeta"

= PROPER (TRIM (A2)) : Ram Seeta

→ lower convert to lowercase, upper convert to upper case: Ram zlower (A3) = ram
I am zupper (A3) = I AM.

→ left 1st character display acc to range
right last character display acc to range.

EXCEL: zleft (A5, 2) : BX

zright (A5, 3) : cel.

→ mid give between char in given range.

EXCEL: mid (A7, 2, 3)

3 char starting from position 2 : xce.

Extract 1st, middle & LAST name:

A1: Alice, Claire, Ben.

1st name: zLEFT (A1, FIND (" ", A1) - 1)

→ FIND (" ", A1) → finds the position of first space
i.e. 6 because space is after Alice.

→ FIND (" ", A1) - 1 → gives the length of 1st name
i.e. 6 - 1 = 5.

→ zleft (A1, 5) → gives first 5 char: Alice.

middle name: =MID (A1, FIND (" ", A1) + 1, FIND (" ", A1, FIND (" ", A1)) - FIND (" ", A1) - 1).

→ FIND (" ", A1) → find position of 1st space: 6

→ FIND (" ", A1) + 1 → 6 + 1 = 7 starts starting from position 7

→ FIND (" ", A1, FIND (" ", A1) + 1) → finds the 2nd space = 13

→ FIND (" ", A1) - 1 → 6 - 1 i.e. 13 - 6 - 1 = 6.

→ So 5 char extract: Claire = MID (A1, 7, 6)

last name: $DI = RIGHT(A1, LEN(A1) - FIND(" ", A1, FIND(" ", A1) + 1))$

- $LEN(A1)$ - length of A1 i.e. $A1 = 16$
- $FIND(" ", A1, FIND(" ", A1) + 1)$ → find position of 2nd space: 13.
- $LEN(A1) - FIND(" ", A1, FIND(" ", A1) + 1)$
→ $16 - 13 = 3$.
- $RIGHT(A1, 3)$: last 3 char: BEN.

Date Handling: Date function returns the sequential numbers that represents a particular date.

Syntax: $DATE(\text{year}, \text{month}, \text{day})$.

→ Default date format: mm/dd/yyyy. ("1/27/2025")

→ Current Date: $=TODAY()$

Current Date & Time: $=NOW()$

Day from Date: $=DAY(A1)$

Month from Date: $=MONTH(A1)$

YEAR from Date: $=YEAR(A1)$.

(create Date using DATE): $=DATE(2025, 8, 12)$.

Years B/w 2 Dates: $DATEDIF("1-Jan-2020", "1-Jan-2025", "Y")$: 5.

month n nt n : $n("n n n", "n n n", "m n")$: 60.

Day n n n : $n("n n n", "n n n", "d n")$: 1826

Time Handling:

- ① $HOUR(\text{Time})$: Extracts hour component from time
- ② $MINUTE(\text{time})$: n minute n n. time give by time
- ③ NOW returns serial number of current date & time.
- ④ $SECOND(\text{time})$: Extract second n n n n n
- ⑤ $TIME(\text{hour}, \text{minute}, \text{second})$ - h:0-23 m&s: 0-59.
- ⑥ $TIMEVALUE(\text{time_text})$: converts a time from text to a serial number.

Add: hours: $n * (1/24)$ Ex: $NOW() + 6 * (1/24)$ (6 hours)

minutes: $n * (1/24/60)$ Ex: $NOW() + 15 * (1/24/60)$ (15 min)

second: $n * (1/24/60/60)$ Ex: $NOW() + 30 * (1/24/60/60)$ (30 sec)

Function	Description	Example	Output
DATE	Returns serial number of specific days.	=DATE(2025, 8, 10)	10-Aug-2025
DATEDIF	Days, months/years b/w 2 dates.	=DATEDIF("1-Jan-25", "10-Aug-25", "Y")	7
DATEVALUE	Convert date text to serial no	=DATEVALUE("10-Aug-2025")	45458
DAY	Day of month from date.	=DAY("10-Aug-2025")	10
DAYS	No. of days b/w 2 dates	=DAYS("10-Aug-2025", "1-Jan-25")	221
DAYS360	Days b/w 2 dates.	=DAYS360("1-Jan-25", "10-Aug-25")	219
EDATE	Date after/before given month.	=EDATE("10-Aug-25", 3)	10-Nov-25
EDOMONTH	Last day of month, before/after	=EDOMONTH("10-Aug-25", 1)	30-Sep-25
HOUR	Extract Hour from time.	=HOUR("14:45")	14
ISOWEEKNUM	ISO week no of date.	=ISOWEEKNUM("10-Aug-25")	32
MINUTE	Extract min from time.	=MINUTE("14:45")	45
MONTH	Month no from date.	=MONTH("10-Aug-25")	8
NETWORKDAYS.INTL	Work days with dates with custom weekends	=NETWORKDAYS.INTL("1-Aug-2025", "10-Aug-2025", 7)	7
Now	Current date & time.	=NOW()	10-Aug-2025 15:30
Second	Extract seconds from Time.	=SECOND("14:45:30")	30
Time	Serial no of specific Time.	=TIME(14, 30, 0)	14:30
TimeValue	Converts timetext to serial no	=TIMEVALUE("2:30 PM")	0.604167
Today	Current date only.	=TODAY()	10-Aug-2025
Weekday	Day of week no	=WEEKDAY("10-Aug-25")	1 - Sunday
Weeknum	Week no in year.	=WEEKNUM("10-Aug-25")	32
workday	Date after workdays after start date	=WORKDAY("1-Aug-25", 5)	8-Aug-25
workday.INTL	Date after workdays with custom weekends.	=WORKDAY.INTL("1-Aug-25", 5, 7)	8-Aug-25
Year	Year from a date	=YEAR("10-Aug-25")	2025
years FRAC	Years fraction b/w 2 dates	=YEARSFRAC("1-Jan-25", "10-Aug-25")	0.60

* Conditional Formatting:

- Highlight cells, create top & bottom rules & apply three types of formatting: Data bars, colors scales & icon sets.
- Apply Conditional formatting based on a result of a formula.
 - If formula TRUE — formatted
 - False — not formatted.
- usually use If with logical function.

Steps:

- ① Select cells to which you want to add conditional formatting
- ② Select Home > Conditional Formatting > New Rule
- ③ Excel displays New Formatting Rule dialog box
- ④ Select use A formula To Determine which cell to format.
- ⑤ In format values where this formula is True range box, type your logical formula.
- ⑥ Select Format to open the Format Cells dialog box.
- ⑦ Use the Number, Font, Border, & Fill tabs to specify formatting you want to apply & then,
- ⑧ Select Ok.

* Sorting & Filtering:

- Sorting: arranging a list of names in alphabetical order, compile a list of product
- Sorting helps to quickly visualize & understand your data better organize & find the data that you want & ultimately make more effective decisions.
- Sorting tool allows you to sort by date, number, alphabetic order & more.

- Filter tool gives the ability to filter a column of data within a table to isolate the key components you need.

* SubTotals with Ranges:

- If you have a list of data that you want to group & summarize you use subtotal & outline to display summary rows or columns.
 - you can create outlines upto 8 levels.
 - This is a quickest way to analyze a range of data.
 - This can only be used on range not on table.
 - Data Sorting is essential.
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